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Deepfake Detection with ResNet18

1. Problem Definition and Motivation

Deepfake technology has created a new set of challenges for online safety, misinformation prevention, and public trust. The goal of this project was to build a working system that can automatically classify individual video frames as real or fake. Even though this is a simplified version of full video deepfake detection, frame classification is an important first step because it allows a model to learn specific visual artifacts that often appear in manipulated content.

The significance of this work comes from the fact that deepfakes are becoming more accessible to create and harder to detect by eye. Automated detection systems can help social media, law enforcement, and digital forensics filter harmful or misleading content. For this project, my main objectives were to preprocess raw videos into usable data, fine tune a CNN model to classify frames, evaluate the model using accepted metrics, and analyze the patterns the model learned. Compared to the midterm, this final project expands the dataset size, improves the preprocessing workflow, and uses a stronger evaluation framework.

2. Related Work and Background

Recent research highlights the growing threat of deepfakes and the need for reliable detection tools. Chesney and Citron describe the real-world risks associated with convincing synthetic media and how it affects political systems, journalism, and online communication. In the field of computer vision, several large scale deepfake benchmarks such as FaceForensics++ and the DeepFake Detection Challenge Dataset have shown that convolutional neural networks and transfer learning are effective approaches for detecting manipulated frames. Rossler et al. demonstrate how CNN based detectors can outperform humans, especially when trained on diverse manipulated examples. Sabir et al. also explore multiple frame-based approaches and found that temporal inconsistencies are not always required for good performance.

The core technical foundation for this project is transfer learning with a pretrained ResNet18 model. ResNet architectures introduced residual blocks, which enable deeper networks without vanishing gradients. Since this project uses limited compute and a moderate dataset size, fine tuning a pretrained CNN is a practical and effective strategy. The novelty in my approach is combining a lightweight frame extraction pipeline with a focused fine-tuning process that reaches high accuracy even with single frame inputs.

References

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Sabir, E., Cheng, J., Jaiswal, A., AbdAlmageed, W., Masi, I., and Natarajan, P. 2019.

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3. Methodology and Implementation

Dataset and Preprocessing

The project used the DeepFake Detection Challenge sample dataset, which contains 400 raw videos with real and fake labels. To meet the final project requirement of at least 5000 training samples, I implemented a uniform frame extraction procedure that saved 16 evenly spaced frames per video. This produced approximately 6400 labeled images. The dataset is imbalanced, with more fake examples than real ones, which is common in deepfake datasets.

Preprocessing steps included:

- reading metadata from JSON
- extracting frames using OpenCV
- organizing images into REAL and FAKE folders
- creating a DataFrame with image paths and labels
- applying normalization based on ImageNet statistics

Model Architecture and Algorithm Selection

I used a ResNet18 model with pretrained ImageNet weights. The final fully connected layer was replaced with a new linear layer for binary classification. Transfer learning was chosen because pretrained CNNs already learn rich features such as edges, textures, lighting patterns, and facial structures. This is helpful for deep-fake detection since manipulated faces often contain distortions around eyes, lips, and skin texture.

Key hyperparameters:

- optimizer: Adam
- learning rate: 0.0001
- batch size: 16
- epochs: 10
- loss function: Cross Entropy Loss
- augmentations: random horizontal flip

Training Process

I split the 6800 images into an 80 percent training set and a 20 percent validation set while maintaining label proportions. Training was done entirely on CPU, which made each epoch slower, but the model was able to converge successfully. Below are the recorded results from 10 epochs:

- train accuracy progressed from 0.810 to 0.958
- validation accuracy peaked around 0.930
- validation loss remained stable and did not show major overfitting

Evaluation Framework

For evaluation, I used:

- accuracy
- confusion matrix
- precision, recall, and f1 score
- visual inspection of correctly and incorrectly classified frames

Accuracy gives a quick overview of performance, while the confusion matrix shows class specific strengths and weaknesses.

4. Results and Analysis

The final model achieved strong performance for a frame-based classifier. The confusion matrix for the 1360 validation frames is shown below. (This refers to the screenshot in your results notebook.)

Validation Accuracy: 92.1 percent

Confusion Matrix:

- True Real predicted as Real: 60
- True Real predicted as Fake: 202
- True Fake predicted as Real: 120
- True Fake predicted as Fake: 978

The model performs very well on fake images and weaker on real ones. This trend shows up in many public deep-fake studies because many real videos have low lighting or compression noise that resembles manipulated patterns.

Precision and recall for the real class were lower, while the fake class scored high across all metrics. The imbalance in training data is likely contributing to this pattern. Even with this issue, the macro average f1 score still indicates that the model generalizes well.

During error analysis, images with unusual lighting, strong shadows, motion blur, or side face angles were more likely to be misclassified. On the other hand, the model reliably detected common deepfake artifacts such as texture inconsistency on cheeks, slightly misaligned facial boundaries, and unnatural sharpness around the mouth.

5. Conclusions and Future Work

This project resulted in a complete deepfake frame classification system that includes preprocessing, model training, evaluation, and analysis. The model reached high accuracy and demonstrated that transfer learning is an effective strategy for this task. The training pipeline was stable, the dataset was expanded to a usable size, and the evaluation tools provided good insight into model behavior.

There are still several limitations. The model only looks at single frames, so it cannot take advantage of temporal inconsistencies across video sequences. The dataset is also imbalanced, and performance on real frames is noticeably lower. Training on a GPU would also open the door to more complex architecture and longer training schedules.

For future improvements, I would explore using a 3D CNN or a recurrent model to incorporate sequences of frames, apply stronger augmentation to balance the real class, and experiment with datasets like FaceForensics++ for more variety. This project also helped me build confidence working with PyTorch, data pipelines, and evaluation tools, which is exactly what I wanted to practice as I work toward becoming stronger in machine learning.